

Uncertainty-Aware Colorization of Astronomical Images: A Monte Carlo Dropout Approach with Perceptual Validation

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Abstract

Composite color images of narrowband astronomical observations (the “Hubble Palette,” mapping SII, H α , and OIII filters to red, green, and blue channels) are produced by deterministic, hand-tuned stretches that give no indication of per-pixel confidence. We present a pipeline that instead learns this mapping with a compact convolutional neural network trained under simulated channel dropout and sensor noise, and that uses Monte Carlo Dropout at inference time to obtain, in addition to a color estimate, a pixel-wise uncertainty map. We apply the method to Hubble Space Telescope WFC3 narrowband observations of M16 (the Eagle Nebula / Pillars of Creation), retrieved from the Mikulski Archive for Space Telescopes (MAST). Qualitatively, the resulting uncertainty map is elevated at saturated stars, frame edges, and cosmic-ray-contaminated pixels, and low across the well-observed nebular body. We further describe a web-based perceptual survey designed to test whether human raters judge model uncertainty consistently with subjective color trustworthiness; we report preliminary results from an initial, small rater pool and outline the data collection still required before a statistically meaningful correlation can be claimed. All code is released publicly to support reproduction and extension to other targets.

Keywords: astronomical image processing — deep learning — uncertainty quantification — Bayesian neural networks — Hubble Space Telescope

1 Introduction

Public-facing astronomical images are almost always composites: multiple monochrome exposures, typically in narrowband filters isolating specific emission lines (e.g., [S II] λ 6716, 6731, H α λ 6563, [O III] λ 5007), are individually stretched and assigned to color channels. The de

facto standard for emission nebulae is the “Hubble Palette” ($\text{SII} \rightarrow \text{R}$, $\text{H}\alpha \rightarrow \text{G}$, $\text{OIII} \rightarrow \text{B}$), popularized by the Hubble Heritage Project. While visually effective, this process is deterministic and offers no mechanism to communicate where the resulting color is well-constrained by the data and where it is not — for example, in regions where one band is missing, saturated, or contaminated by cosmic rays.

Bayesian deep learning offers a practical route to per-pixel confidence estimates without the computational cost of a full Bayesian neural network. `gal2016dropout` showed that a network trained with Dropout can be interpreted as an approximation to a deep Gaussian process, and that keeping Dropout *active* at inference time and averaging multiple stochastic forward passes (Monte Carlo Dropout, MC-Dropout) yields both a predictive mean and a predictive variance, the latter serving as an uncertainty estimate. MC-Dropout has been applied broadly in medical imaging and autonomous driving, where it is important to know not just what a model predicts but how much to trust that prediction; its application to astronomical image colorization, to our knowledge, is not established in the mainstream literature, motivating the exploratory pipeline described here.

This paper makes three contributions: (1) an end-to-end, reproducible pipeline for MC-Dropout-based uncertainty-aware colorization of multi-band astronomical imagery, from raw MAST retrieval through registration, training, and tiled inference; (2) a qualitative analysis of the resulting uncertainty map on a real HST dataset of M16; and (3) the design (and preliminary results) of a perceptual survey intended to test whether the model’s uncertainty is perceptually meaningful to human observers.

2 Data

We use narrowband imaging of M16 (NGC 6611, the Eagle Nebula) obtained with the Wide Field Camera 3 (WFC3/UVIS) aboard the Hubble Space Telescope, retrieved via `astroquery.mast` `astroquery`, `astropy2018`. Three filters were used, each isolating a different ionic emission line:

- **F502N** ([O III], $\lambda 5007$)
- **F656N** ($\text{H}\alpha$, $\lambda 6563$)
- **F673N** ([S II], $\lambda 6716, 6731$)

The OIII and SII exposures originate from the same observing proposal (proposal ID 13926) and share a common pointing. No $\text{H}\alpha$ exposure from the same proposal was available in the archive; the closest-coverage $\text{H}\alpha$ observation (proposal 16359) was retrieved instead. Because its footprint does not fully coincide with the OIII/SII pointing, the three bands were reprojected onto a common World Coordinate System (WCS) grid using `reproject.reproject_interp` `reproject`, and the image was cropped to the largest axis-aligned rectangle with 100% valid coverage in all three bands, found via a maximal-rectangle-in-binary-matrix search over the union-validity mask. This crop is necessary because the naive bounding box of the valid-pixel mask still contains invalid corners when the input footprints are mutually rotated.

3 Methods

3.1 Baseline colorization

As a reference, we construct a deterministic Hubble Palette composite: each registered, cropped band is independently stretched with a 1st–99.5th percentile linear stretch and assigned to its corresponding RGB channel. This baseline has no notion of confidence and serves only as a visual and numerical reference point for the learned model.

3.2 MC-Dropout network

We use a compact fully convolutional network (four convolutional layers, 32–64 channels, ReLU activations, 1×1 output convolution with a sigmoid nonlinearity) with `Dropout2d` ($p = 0.3$) inserted after each hidden layer. Because no independently verified “true color” exists for these data, we cannot train a supervised colorization model in the conventional sense. Instead we frame the learning problem as *reconstruction under simulated data degradation*: during training, each input patch has, with probability 0.4, one of its three bands set to zero (simulating a missing/uncovered band, as encountered with the real H α footprint mismatch), plus additive Gaussian noise ($\sigma = 0.05$) applied to all channels. The training target is the clean, undegraded three-band patch. This framing gives the resulting uncertainty a concrete operational meaning: it should be elevated wherever the network is extrapolating from incomplete or noisy information, which is directly relevant to the missing-coverage problem observed in Section 2.

The network is trained with Adam (learning rate 10^{-3}) and an MSE loss on 128×128 patches randomly sampled from the registered, percentile stretched bands, for 30 epochs (4000 patches/epoch, batch size 16), on a single NVIDIA RTX 4060 laptop GPU.

3.3 Uncertainty inference

At inference, Dropout is kept active (`model.train()` mode) and the full image is processed in overlapping 256×256 tiles (stride 192) to avoid seam artifacts. Each tile is passed through the network $N = 30$ times; the per-pixel mean across passes is taken as the color estimate and the per-pixel standard deviation as the uncertainty estimate. Tiles are recombined by averaging in overlap regions.

3.4 Perceptual validation survey

To assess whether the resulting uncertainty map corresponds to human judgments of color trustworthiness, we selected 14 180×180 -pixel patches spanning the observed range of mean per-patch uncertainty (from the lowest to the highest decile), with display order randomized per session. Human raters, blind to the underlying uncertainty values, scored each patch on a 5-point scale (1 = “completely artificial,” 5 = “completely natural and trustworthy”). The instrument is a self-contained, browser-based survey (Supplementary material; [survey/uncertainty_survey.html](#)) that requires no server-side infrastructure. We planned

to correlate the mean human rating per patch against the model’s mean uncertainty for that patch using Spearman’s rank correlation coefficient ρ .

4 Results

4.1 Registration and baseline

After reprojection, the OIII and SII bands (originating from the same pointing) retained 93% pixel coverage relative to the reference frame, while the independently-pointed H α band retained only 26–33% coverage depending on the specific archival exposure tested. The maximal-rectangle crop yields a final working image of 1920×3873 pixels with 100% valid coverage in all three bands.

4.2 Uncertainty map

Qualitatively, the MC-Dropout uncertainty map (Figure 1, not reproduced here; see **survey/** outputs) shows three consistent patterns: (i) elevated uncertainty at a saturated foreground star, (ii) elevated uncertainty along image borders, where the network has reduced spatial context, and (iii) elevated uncertainty at scattered single-pixel and short-streak features consistent with cosmic-ray hits, which by construction appear in only one of the three independently-exposed bands and therefore constitute exactly the kind of “missing/inconsistent information” the training procedure was designed to flag. The bulk of the nebular body, where all three bands are well exposed and mutually consistent, shows low uncertainty.

4.3 Preliminary perceptual survey results

At the time of writing, $n = 3$ raters had completed the survey (partial completion permitted). The Spearman correlation between mean human rating and model uncertainty across the 14 patches was $\rho \approx 0.00$ ($p \approx 1.0$), i.e., no significant relationship was detected. Given $n = 3$, this is expected to be underpowered and should not be interpreted as evidence against the method; a minimum of 10–15 raters is planned before drawing conclusions. One qualitative observation is noteworthy: the single patch with the highest model uncertainty (containing the saturated star) also showed the largest disagreement *among raters* (ratings of 1, 4, and 4 on the 5-point scale), suggesting that this patch was subjectively ambiguous to humans as well, even though the average rating alone does not yet show a monotonic trend with uncertainty. Data collection is ongoing; this section will be updated with a full statistical analysis (correlation, inter-rater agreement, and a scatter plot of rating vs. uncertainty) once a sufficient sample is collected.

5 Discussion

Several limitations should be noted. First, because no ground-truth color exists, the network’s training target is itself a deterministic stretch of the observed bands rather than

an independently verified “correct” color; the uncertainty we estimate is therefore best understood as *data-completeness/consistency uncertainty* (how much the network relies on extrapolation from degraded input) rather than a calibrated statistical confidence interval on true physical flux. Second, the perceptual survey, as of this draft, has too few respondents for a statistically meaningful correlation; the survey is designed to scale to additional raters without further engineering (see `07_generate_survey_patches.py` and the accompanying HTML instrument), and this is the immediate next step in the project. Third, results are demonstrated on a single target (M16); generalization to other nebulae, filter sets, and instruments (e.g., JWST NIRCам) remains to be tested.

6 Conclusion and Future Work

We have demonstrated a complete, reproducible pipeline for MC-Dropout-based uncertainty-aware colorization of multi-band Hubble Space Telescope imagery, including automated data retrieval, robust registration under mismatched fields of view, a degradation-aware training scheme that gives the resulting uncertainty a concrete operational interpretation, and a lightweight perceptual survey instrument for human validation. Immediate next steps are (1) completing perceptual data collection to a statistically adequate sample size, (2) extending the pipeline to additional targets and instruments, and (3) exploring alternative uncertainty-estimation techniques (deep ensembles, evidential deep learning) as a comparison point for MC-Dropout. All code, the survey instrument, and this manuscript source are released publicly (see repository).

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